

Algebra II with Trigonometry

Chapter 5.6

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:

[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Problems

1. In $\triangle ABC$, the measure of $\angle C$ is 60° . The side lengths a and b differ by exactly 3, and side c has a length of 7. Determine the lengths of sides a and b .
2. A triangle has side lengths $13x$, $14x$, and $15x$ for some positive real number x . If its area is numerically equal to its semiperimeter, determine the exact value of x .
3. In quadrilateral $ABCD$, the side lengths are $AB = 7$, $BC = 24$, $CD = 20$, and $DA = 15$. Given that $\angle ABC$ is exactly 90° , determine the measure of $\angle ADC$ without writing down any calculations.
4. A ship leaves port and sails 40 miles at a bearing of 020° (measured clockwise from due North). It then turns and sails another 40 miles at a bearing of 140° . What is the ship's exact straight-line distance from the port?
5. The side lengths a , b , and c of a triangle satisfy the algebraic relation $2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4) = 576$. Mentally determine the area of the triangle.
6. The side lengths of an oblique triangle are given by the expressions $2x + 1$, $x^2 - 1$, and $x^2 + x + 1$ for some integer $x > 1$. Determine the degree measure of the largest angle in this triangle.